Missing Steps and Extra Workings for PDEs by Evans

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Chapter 5

Example 4. *Let $\{r_n\}_{n\in\mathbb{N}}$ be a dense subset of $B_1(\mathbf{0}) \subset \mathbb{R}^d$, the open unit ball centred at the origin. Write*

$$u(x) = \sum_{n\in\mathbb{N}} \frac{1}{2^n}|x - r_n|^{-\alpha} \quad (x \in B_1(\mathbf{0})).$$

Then $u \in W^{1,p}(B_1(\mathbf{0}))$ if $0 < \alpha < \frac{n-p}{p}$.

Proof. In example 3 of the same chapter we see that $v \in W^{1,p}(B_1(\mathbf{0}))$ given by $v(x) = |x|^{-\alpha}$ if $0 < \alpha < \frac{d-p}{p}$. We proceed by decomposing u as the limit of partial sums and define

$$u^N(x) = \sum_{n=1}^{N} \frac{1}{2^n}|x - r_n|^{-\alpha}.$$

I claim $(u^N)_{N\in\mathbb{N}}$ is a Cauchy sequence in $W^{1,p}(B_1(\mathbf{0}))$.

$$\begin{aligned}
\|u^N\|_{W^{1,p}(B_1(\mathbf{0}))} &\leq \sum_{n=1}^{N} \frac{1}{2^n} \left\|\left(x \mapsto |x - r_n|^{-\alpha}\right)\right\|_{W^{1,p}(B_1(\mathbf{0}))} \\
&= \sum_{n=1}^{N} \frac{1}{2^n} \left[\int_{B_1(\mathbf{0})\setminus\{r_n\}} |v(x - r_n)|^p + |\nabla v(x - r_n)|^p \, dx\right]^{\frac{1}{p}} \\
&= \sum_{n=1}^{N} \frac{1}{2^n} \left[\int_{B_1(\mathbf{0})\setminus\{r_n\}-r_n} |v(y)|^p + |\nabla v(y)|^p \, dy\right]^{\frac{1}{p}} \\
&\leq \sum_{n=1}^{N} \frac{1}{2^n} \left[\int_{B_2(\mathbf{0})\setminus\{r_n\}} |v(y)|^p + |\nabla v(y)|^p \, dy\right]^{\frac{1}{p}} \\
&\leq \sum_{n=1}^{N} \frac{1}{2^n} \|v\|_{W^{1,p}(B_2(\mathbf{0}))} \\
&\lesssim \sum_{n=1}^{N} \frac{1}{2^n}.
\end{aligned}$$

Where we have used a change of variables and that the translated unit ball is contained within an open ball of radius 2 since $r_n \in [0, 1)$ for each $n \in \mathbb{N}$. By example 3 we can drop the dependence on the norm of v when $0 < \alpha < \frac{n-p}{p}$, it then follows that $(u^N)_{N\in\mathbb{N}}$ is a Cauchy sequence in $W^{1,p}(B_1(\mathbf{0}))$ and since $W^{1,p}(B_1(\mathbf{0}))$ is a Banach space there exists a $w \in W^{1,p}(B_1(\mathbf{0}))$ such that $u^N \to w$ as $N \to \infty$ with respect to $\|\cdot\|_{W^{1,p}(B_1(\mathbf{0}))}$. We can extract a subsequence such that u^{N_j} converges point-wise to w for almost every $x \in B_1(0) \setminus \{r_n\}_{n\in\mathbb{N}}$, we also have point-wise convergence almost everywhere to u so u and w are equal almost everywhere. Therefore $u \in W^{1,p}(B_1(\mathbf{0}))$. $\square$

Chapter 6

Theorem 1 (Lax-Milgram)**.** *Let $B : H \times H \to \mathbb{R}$ be a bilinear form such that there exists $\alpha, \beta > 0$ satisfying*

1. $|B[u, v]| \leq \alpha \|u\| \|v\| \quad (u, v \in H)$
2. $\beta \|u\|^2 \leq B[u, y] \quad (u \in H)$.

Then for $f \in H^*$ there exists a unique $u \in H$ such that

$$B[u, v] = \langle f, v \rangle,$$

for all $v \in H$.

Proof. Fix $u \in H$, the mapping $v \mapsto B[u, v]$ is a bounded linear functional so by the Riesz representation theorem there exists a unique $w \in H$ such that

$$B[u, v] = (w, v),$$

for all $v \in H$. We endeavor to construct an operator $A : H \to H$ that maps u to w and write $Au = w$ when the above equality is satisfied. First we claim that A is bounded and linear, linearity is immediate using $B[u, v] = (Au, v)$. For boundedness we compute

$$\|Au\|^2 = (Au, Au) = B[u, Au] \leq \alpha \|u\| \|Au\|,$$

which proves the claim. Next we claim that A is injective and its image is closed in H, indeed

$$\beta \|u\|^2 \leq B[u, u] = (Au, u) \leq \|Au\| \|u\|,$$

which implies that $Au = 0 \iff u = 0$ hence injectivity. We use the same computation to prove the image of A is closed, let $(v_n)_{n \in \mathbb{N}}$ be a sequence in $\mathrm{Im}(A)$ converging to $v \in H$. By injectivity we find a sequence $(u_n)_{n \in \mathbb{N}}$ in H such that $Au_n = v_n$. Using the previous computation

$$\beta \|u_n - u_m\| \leq \|A(u_n - u_m)\| = \|v_n - v_m\| \to 0,$$

as $n, m \to \infty$ and hence $(u_n)_{n \in \mathbb{N}}$ is Cauchy in H so converges to an element $u \in H$. It remains to show $Au = v$

$$\|v - Au\| \leq \|v - v_n\| + \|v_n - Au\| = \|v - v_n\| + \|A\| \|u_n - u\| \to 0,$$

as $n \to \infty$, therefore $\mathrm{Im}(A)$ is closed in H. Next we claim that $\mathrm{Im}(A) = H$, since $\mathrm{Im}(A)$ is a closed linear subspace of H, we can decompose it into orthogonal complements $H = \mathrm{Im}(A) \oplus \mathrm{Im}(A)^\perp$. By contradiction suppose that $\mathrm{Im}(A) \neq H$ then there exists $w \in H \setminus \{\mathbf{0}\}$ with $w \in \mathrm{Im}(A)^\perp$

$$\beta \|w\| \leq B[w, w] = (Aw, w) = 0,$$

which contradict $w \neq \mathbf{0}$. Let $f \in H^*$, we apply the Riesz representation theorem to find some $w \in H$ such that

$$\langle f, v \rangle = (w, v),$$

for all $v \in H$. By construction we can find $u \in H$ such that $Au = w$ and see that

$$B[u, v] = (Au, v) = (w, v) = \langle f, v \rangle,$$

for all $v \in H$. Finally, we show that the choice of $u \in H$ is unique, suppose there exist u and $\tilde{u}$ satisfying the above equality then $B[u - \tilde{u}, v] = 0$ for all $v \in H$. Taking $v = u - \tilde{u}$ and applying the second assumption on B

$$\beta \|u - \tilde{u}\|^2 \leq B[u - \tilde{u}, u - \tilde{u}] = 0,$$

therefore $u = \tilde{u}$. □

Theorem 2 (Energy estimates)**.** *There exists constants $\alpha, \beta > 0$ and $\gamma \geq 0$ such that*

1. $|B[u, v]| \leq \alpha \|u\|_{H_0^1} \|v\|_{H_0^1}$
2. $\beta \|u\|_{H_0^1}^2 \leq B[u, v] + \gamma \|u\|_{L^2}^2$

for all $u, v \in H_0^1$.

Proof. Continuity is readily checked

$$
\begin{aligned}
|B[u,v]| &\leq \sum_{i,j=1}^{n} \|a_{i,j}\|_{L^\infty} \int_U |Du||Dv| + \sum_{i=1}^{n} \|b_i\|_{L^\infty} \int_U |Du||v| + \|c\|_{L^\infty} \int_U |u||v| \\
&\leq \|Du\|_{L^2}\|Dv\|_{L^2} \sum_{i,j=1}^{n} \|a_{i,j}\|_{L^\infty} + \|Du\|_{L^2}\|v\|_{L^2} \sum_{i=1}^{n} \|b_i\|_{L^\infty} + \|c\|_{L^\infty}\|u\|_{L^2}\|v\|_{L^2} \\
&\leq \left[\sum_{i,j=1}^{n} \|a_{i,j}\|_{L^\infty} + \sum_{i=1}^{n} \|b_i\|_{L^\infty} + \|c\|_{L^\infty} \right] \left[\|Du\|_{L^2} + \|u\|_{L^2}\right] \left[\|Dv\|_{L^2} + \|v\|_{L^2}\right] \\
&\leq C^2 \left[\sum_{i,j=1}^{n} \|a_{i,j}\|_{L^\infty} + \sum_{i=1}^{n} \|b_i\|_{L^\infty} + \|c\|_{L^\infty} \right] \|u\|_{H_0^1}\|v\|_{H_0^1}.
\end{aligned}
$$

Where we have used Cauchy-Schwarz and that the norm given by $\|u\|_* = \|Du\|_{L^2} + \|u\|_{L^2}$ is equivalent to $\|\cdot\|_{H_0^1}$ and $C > 0$ such that $\|u\|_* \leq \|u\|_{H_0^1}$.

Furthermore, by the ellipticity of L there exists $\theta > 0$ such that $\xi^\intercal A(x)\xi \geq \theta|\xi|^2$ for almost every $x \in U$ and every $\xi \in \mathbb{R}^n$. Applying this pointwise with $\xi = Du(x)$ and integrating over U

$$
\begin{aligned}
\theta \int_U |Du|^2 &\leq \int_U \sum_{i,j=1}^{n} a_{i,j} u_{x_i} u_{x_j} \\
&= B[u,u] - \int_U \sum_{i=1}^{n} b_i u_{x_i} u + cu^2 \\
&\leq B[u,u] + \sum_{i=1}^{n} \|b_i\|_{L^\infty} \int_U |Du||u| + \|c\|_{L^\infty} \int_U |u|^2.
\end{aligned}
$$

Applying Cauchy's inequality with $\varepsilon > 0$ to the middle term gives

$$
\int_U |Du||u| \leq \varepsilon \|Du\|_{L^2}^2 + \frac{1}{4\varepsilon}\|u\|_{L^2}^2,
$$

choosing ε small enough so that $\varepsilon \sum_{i=1}^{n} \|b_i\|_{L^\infty} < \frac{\theta}{2}$

$$
\begin{aligned}
\theta \int_U |Du|^2 &\leq B[u,u] + \frac{\theta}{2}\|Du\|_{L^2}^2 + \left[\frac{1}{4\varepsilon} \sum_{i=1}^{n} \|b_i\|_{L^\infty} + \|c\|_{L^\infty} \right] \|u\|_{L^2}^2 \\
\frac{\theta}{2}\|Du\|_{L^2}^2 &\leq B[u,u] + \left[\frac{1}{4\varepsilon} \sum_{i=1}^{n} \|b_i\|_{L^\infty} + \|c\|_{L^\infty} \right] \|u\|_{L^2}^2.
\end{aligned}
$$

Adding $\frac{\theta}{2}\|u\|_{L^2}^2$ to both sides yields the result. $\square$